Recursive Functions, Relations and Their Properties

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Recursive Functions

What is Recursion?

Idea

A function can be defined in terms of **its previous values**.

- We define some **starting values** directly.
- We give a **rule** that tells us how to compute the next value from earlier ones.
- Then we can generate as many values of the function as we want.

Recursively Defined Functions

We use two steps to define a function with the set of nonnegative integers as its domain:

BASIS STEP

Specify the value of the function at zero (and sometimes also at one, two, etc.).

RECURSIVE STEP

Give a rule for finding its value at an integer from its values at smaller integers.

- The recursive step must always move **backward** to already-known values.
- This guarantees we can actually compute $f(n)$ for any n.

Example 1

Problem 1

Find $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if f is defined recursively by

$$f(0) = -1, \quad f(1) = 2,$$

$$f(n+1) = f(n) + 3f(n-1)$$

Given

$$f(0) = -1, \quad f(1) = 2, \quad f(n+1) = f(n) + 3f(n-1).$$

Compute $f(2)$ using $n = 1$:

$$f(2) = f(1) + 3f(0) = 2 + 3(-1) = 2 - 3 = -1.$$

Example 1

Compute $f(3)$ using $n = 2$:

$$f(3) = f(2) + 3f(1) = (-1) + 3(2) = -1 + 6 = 5.$$

Compute $f(4)$ using $n = 3$:

$$f(4) = f(3) + 3f(2) = 5 + 3(-1) = 5 - 3 = 2.$$

Compute $f(5)$ using $n = 4$:

$$f(5) = f(4) + 3f(3) = 2 + 3(5) = 2 + 15 = 17.$$

Final values:

$$f(2) = -1, \quad f(3) = 5, \quad f(4) = 2, \quad f(5) = 17.$$

Example 2

Problem 2

Find $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if f is defined recursively by

$$f(0) = -1, \quad f(1) = 2,$$

$$f(n+1) = f(n)^2 f(n-1).$$

Given

$$f(0) = -1, \quad f(1) = 2, \quad f(n+1) = f(n)^2 f(n-1).$$

Compute $f(2)$ using $n = 1$:

$$f(2) = f(1)^2 f(0) = (2)^2 \cdot (-1) = 4 \cdot (-1) = -4.$$

Compute $f(3)$ using $n = 2$:

$$f(3) = f(2)^2 f(1) = (-4)^2 \cdot 2 = 16 \cdot 2 = 32.$$

Example 2

Compute $f(4)$ using $n = 3$:

$$f(4) = f(3)^2 f(2) = (32)^2 \cdot (-4).$$

Since $(32)^2 = 1024$,

$$f(4) = 1024 \cdot (-4) = -4096.$$

Compute $f(5)$ using $n = 4$:

$$f(5) = f(4)^2 f(3) = (-4096)^2 \cdot 32.$$

Now $(-4096)^2 = 4096^2 = 16{,}777{,}216$, and

$$16{,}777{,}216 \times 32 = 536{,}870{,}912.$$

Final values:

$$f(2) = -4, \quad f(3) = 32, \quad f(4) = -4096, \quad f(5) = 536{,}870{,}912.$$

Example 3

Problem 3

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if $f(n)$ is defined recursively by

$$f(0) = 3,$$

$$f(n+1) = -2f(n).$$

Example 3

Given

$$f(0) = 3, \quad f(n+1) = -2f(n).$$

Compute:

$$f(1) = -2f(0) = -2 \cdot 3 = -6,$$
$$f(2) = -2f(1) = -2 \cdot (-6) = 12,$$
$$f(3) = -2f(2) = -2 \cdot 12 = -24,$$
$$f(4) = -2f(3) = -2 \cdot (-24) = 48,$$
$$f(5) = -2f(4) = -2 \cdot 48 = -96.$$

Final values:

$$f(1) = -6, \quad f(2) = 12, \quad f(3) = -24, \quad f(4) = 48, \quad f(5) = -96.$$

Example 4

Problem 4

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if $f(n)$ is defined recursively by

$$f(0) = 3,$$

and for $n = 0, 1, 2, \ldots$

$$f(n+1) = 3f(n) + 7.$$

We again start from $f(0) = 3$ and apply the recursive definition one step at a time.

Example 4

Given

$$f(0) = 3, \quad f(n+1) = 3f(n) + 7.$$

Compute:

$$f(1) = 3f(0) + 7 = 3 \cdot 3 + 7 = 9 + 7 = 16,$$
$$f(2) = 3f(1) + 7 = 3 \cdot 16 + 7 = 48 + 7 = 55,$$
$$f(3) = 3f(2) + 7 = 3 \cdot 55 + 7 = 165 + 7 = 172,$$
$$f(4) = 3f(3) + 7 = 3 \cdot 172 + 7 = 516 + 7 = 523,$$
$$f(5) = 3f(4) + 7 = 3 \cdot 523 + 7 = 1569 + 7 = 1576.$$

Final values:

$$f(1) = 16, \quad f(2) = 55, \quad f(3) = 172, \quad f(4) = 523, \quad f(5) = 1576.$$

Example 5

Problem 5

Suppose that f is defined recursively by

$$f(0) = 3, \qquad f(n+1) = 2f(n) + 3.$$

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

In this version:

- We **rewrite** $f(n)$**,** $f(n-1)$**,** $f(n-2)$**, ... all the way back to** $f(0)$
- to get a formula for $f(n)$ in terms of $f(0)$ only.

Example 5 (Using the Closed Form)

We start from the recurrence in a convenient index form:

$$f(n) = 2f(n-1) + 3 \quad \text{for } n \geq 1, \qquad f(0) = 3.$$

Now substitute *recursively*:
First step:

$$f(n) = 2f(n-1) + 3.$$

Replace $f(n-1)$ using the same rule:

$$f(n-1) = 2f(n-2) + 3.$$

So

$$f(n) = 2\big(2f(n-2) + 3\big) + 3 = 2^2 f(n-2) + 2 \cdot 3 + 3.$$

Replace $f(n-2)$:

$$f(n-2) = 2f(n-3) + 3.$$

Then

$$\begin{aligned} f(n) &= 2^2\big(2f(n-3) + 3\big) + 2 \cdot 3 + 3 \\ &= 2^3 f(n-3) + 2^2 \cdot 3 + 2 \cdot 3 + 3. \end{aligned}$$

Example 5 (Using the Closed Form)

If we keep doing this k times, we get the pattern

$$f(n) = 2^k f(n-k) + 3\big(2^{k-1} + 2^{k-2} + \cdots + 2 + 1\big).$$

Now choose $k = n$, so that $n - k = 0$:

$$f(n) = 2^n f(0) + 3\big(2^{n-1} + 2^{n-2} + \cdots + 2 + 1\big).$$

We know $f(0) = 3$, so

$$f(n) = 2^n \cdot 3 + 3\big(2^{n-1} + 2^{n-2} + \cdots + 2 + 1\big).$$

The sum in parentheses is a geometric series:

$$2^{n-1} + 2^{n-2} + \cdots + 2 + 1 = 2^n - 1.$$

Therefore,

$$f(n) = 3 \cdot 2^n + 3(2^n - 1) = 3 \cdot 2^n + 3 \cdot 2^n - 3 = 6 \cdot 2^n - 3.$$

$$\boxed{f(n) = 6 \cdot 2^n - 3}$$

Example 5 (Using the Closed Form)

Now we can evaluate using

$$f(n) = 6 \cdot 2^n - 3.$$

$$f(1) = 6 \cdot 2^1 - 3 = 12 - 3 = 9$$
$$f(2) = 6 \cdot 2^2 - 3 = 24 - 3 = 21$$
$$f(3) = 6 \cdot 2^3 - 3 = 48 - 3 = 45$$
$$f(4) = 6 \cdot 2^4 - 3 = 96 - 3 = 93$$

Final values:

$$\boxed{f(1) = 9, \quad f(2) = 21, \quad f(3) = 45, \quad f(4) = 93}$$

Relations on Sets

Definition: Binary Relation

Definition

Let A and B be sets. A **binary relation** from A to B is a **subset** of $A \times B$.

- Elements of a relation are ordered pairs (a, b) with $a \in A$, $b \in B$.
- Saying $a\,R\,b$ means $(a, b) \in R$.

Definition

A **relation on a set** A is a relation from A to A (so it's a subset of $A \times A$).

Example: Relation from A to B

Problem

Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Consider the relation

$$R = \{(0, a), (0, b), (1, a), (2, b)\}.$$

- $(0, a) \in R$, so we say $0 \, R \, a$.
- $(1, b) \notin R$, so $1 \not R b$.

We can show this relation in two equivalent ways:

- Arrow diagram (graphical)
- Relation table (matrix-style)

Tabular Representation of a Relation

Table for R

We make a table with rows from A and columns from B. We mark a cell if that ordered pair is in R.

R	a	b
0	$\times$	$\times$
1	$\times$	
2		$\times$

- Row x, column y has a mark if $(x, y) \in R$.
- For instance, row 1 / column a is marked, so $(1, a) \in R$.
- Row 1 / column b is empty, so $(1, b) \notin R$.

Example: Relation on One Set

Problem

Let $A = \{1, 2, 3, 4\}$. Consider the relation

$$R = \{(a, b) \mid a \text{ divides } b\}.$$

Which ordered pairs are in R?

Solution

Because $(a, b) \in R$ iff $a \mid b$ and both $a, b \in \{1, 2, 3, 4\}$, we list:

$$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

- Example: $2R4$ because $2 \mid 4$.
- But $(3, 4) \notin R$ because $3 \nmid 4$.

Example: How Many Relations Are Possible?

Problem

How many relations are there on a set with n elements?

Solution

A relation on a set A is any subset of $A \times A$.

- If $|A| = n$, then $|A \times A| = n^2$.
- A set with m elements has 2^m subsets.

So the number of possible relations on A is

$$2^{n^2}.$$

Example: If $A = \{a, b, c\}$, then $n = 3$ and there are

$$2^{3^2} = 2^9 = 512$$

possible relations on A.

Reflexive Relation

Definition

A relation R on a set A is called **reflexive** if and only if

$$(a, a) \in R \quad \text{for every } a \in A.$$

- Every element of A must relate to itself.
- In the table of the relation, all diagonal entries must be marked.

Example

If $A = \{1, 2, 3, 4\}$, the relation

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

is reflexive.

Example 1: $R_1 = \{(a, b) \mid a = b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_1 : A \to A = \{(a, b) \mid a = b\}$. Determine whether R_1 is reflexive and show its tabular representation.

Solution

$$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

R_1	1	2	3	4
1	$\times$			
2		$\times$		
3			$\times$	
4				$\times$

Conclusion: All diagonal pairs are present. Hence R_1 is **reflexive**.

Example 2: $R_2 = \{(a, b) \mid a \neq b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_2 : A \to A = \{(a, b) \mid a \neq b\}$. Determine whether R_2 is reflexive and show its tabular representation.

Solution

All pairs where a and b differ:

$$R_2 = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}.$$

R_2	1	2	3	4
1		×	×	×
2	×		×	×
3	×	×		×
4	×	×	×	

Conclusion: None of the diagonal pairs (a, a) are present, so R_2 is **not reflexive**.

Example 3: $R_3 = \{(a, b) \mid a \mid b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_3 : A \to A = \{(a, b) \mid a \text{ divides } b\}$. Determine whether R_3 is reflexive and show its tabular representation.

Solution

$$R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

R_3	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Conclusion: Each element divides itself, so R_3 is **reflexive**.

Example 4: $R_4 = \{(a, b) \mid a < b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_4 : A \to A = \{(a, b) \mid a < b\}$. Determine whether R_4 is reflexive and show its tabular representation.

Solution

$$R_4 = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}.$$

R_4	1	2	3	4
1		×	×	×
2			×	×
3				×
4				

Conclusion: No diagonal elements are included, so R_4 is **not reflexive**.

Example 5: $R_5 = \{(a, b) \mid ab < 10\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_5 : A \to A = \{(a, b) \mid ab < 10\}$. Determine whether R_5 is reflexive and show its tabular representation.

Solution

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,1), (4,2)\}.$$

R_5	1	2	3	4
1	×	×	×	×
2	×	×	×	×
3	×	×	×	
4	×	×		

Conclusion: Diagonal pairs (a, a) exist for $a = 1, 2, 3$, but not for $a = 4$. Hence R_5 is **not fully reflexive**.

Irreflexive Relation

Definition

A relation R on a set A is called **irreflexive** if and only if

$$(a, a) \notin R \quad \text{for every } a \in A.$$

- No element of A is related to itself.
- In the table of the relation, **all diagonal entries must be empty**.

Example

If $A = \{1, 2, 3, 4\}$, the relation

$$R = \{(1, 2), (2, 1), (2, 3), (3, 4)\}$$

is irreflexive, because it does *not* contain $(1, 1), (2, 2), (3, 3), (4, 4)$.

Example 1: $R_1 = \{(a, b) \mid a = b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_1 : A \to A = \{(a, b) \mid a = b\}$. Determine whether R_1 is irreflexive.

Solution

All pairs with $a = b$:

$$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

R_1	1	2	3	4
1	×			
2		×		
3			×	
4				×

Conclusion: Every diagonal pair (a, a) *is* in R_1. So R_1 is **not irreflexive**.

Example 2: $R_2 = \{(a, b) \mid a \neq b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_2 : A \to A = \{(a, b) \mid a \neq b\}$. Determine whether R_2 is irreflexive.

Solution

All ordered pairs where the two entries are different:

$$R_2 = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}.$$

R_2	1	2	3	4
1		×	×	×
2	×		×	×
3	×	×		×
4	×	×	×	

Conclusion: None of $(1, 1), (2, 2), (3, 3), (4, 4)$ appear. So R_2 **is irreflexive**.

Example 3: $R_3 = \{(a, b) \mid a \mid b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_3 : A \to A = \{(a, b) \mid a \text{ divides } b\}$. Determine whether R_3 is irreflexive.

Solution

We include (a, b) when a divides b:

$$R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

R_3	1	2	3	4
1	$\times$	$\times$	$\times$	$\times$
2		$\times$		$\times$
3			$\times$	
4				$\times$

Conclusion: All diagonal pairs $(1,1), (2,2), (3,3), (4,4)$ are present, so R_3 is **not irreflexive**.

Example 4: $R_4 = \{(a, b) \mid a < b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_4 : A \to A = \{(a, b) \mid a < b\}$. Determine whether R_4 is irreflexive.

Solution

We include (a, b) when a is strictly less than b:

$$R_4 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

R_4	1	2	3	4
1		×	×	×
2			×	×
3				×
4				

Conclusion: No diagonal pairs (a, a) appear, so R_4 **is irreflexive**.

Example 5: $R_5 = \{(a, b) \mid ab < 10\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_5 : A \to A = \{(a, b) \mid ab < 10\}$. Determine whether R_5 is irreflexive.

Solution

List all (a, b) with $a \times b < 10$:

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,1), (4,2)\}.$$

R_5	1	2	3	4
1	×	×	×	×
2	×	×	×	×
3	×	×	×	
4	×	×		

Conclusion: $(1,1), (2,2), (3,3)$ are in R_5, so diagonal entries are *not* all empty. Therefore R_5 is **not irreflexive**.

Symmetric Relation

Definition

A relation R on a set A is called **symmetric** if and only if

$$(a, b) \in R \implies (b, a) \in R \quad \text{for all } a, b \in A.$$

- Whenever a is related to b, b must also be related to a.
- In the table of the relation, **the matrix is symmetric about the main diagonal**.

Example

If $A = \{1, 2, 3, 4\}$, the relation

$$R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$$

is symmetric, because whenever (a, b) is in R, the reverse pair (b, a) is also in R.

Example 1: $R_1 = \{(a, b) \mid a = b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_1 : A \to A = \{(a, b) \mid a = b\}$. Determine whether R_1 is symmetric.

Solution

All pairs have $a = b$:

$$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

R_1	1	2	3	4
1	×			
2		×		
3			×	
4				×

Conclusion: Each (a, b) automatically satisfies (b, a) since $a = b$. Therefore R_1 is **symmetric**.

Example 2: $R_2 = \{(a, b) \mid a \neq b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_2 : A \to A = \{(a, b) \mid a \neq b\}$. Determine whether R_2 is symmetric.

Solution

All ordered pairs where a and b differ:

$$R_2 = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}.$$

R_2	1	2	3	4
1		×	×	×
2	×		×	×
3	×	×		×
4	×	×	×	

Conclusion: Whenever (a, b) appears, (b, a) also appears (since inequality is mutual). Hence R_2 is **symmetric**.

Example 3: $R_3 = \{(a, b) \mid a \mid b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_3 : A \to A = \{(a, b) \mid a \text{ divides } b\}$. Determine whether R_3 is symmetric.

Solution

We include (a, b) when a divides b:

$$R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

R_3	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Conclusion: For example, $(1, 2) \in R_3$ but $(2, 1) \notin R_3$. Therefore R_3 is **not symmetric**.

Example 4: $R_4 = \{(a, b) \mid a < b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_4 : A \to A = \{(a, b) \mid a < b\}$. Determine whether R_4 is symmetric.

Solution

We include (a, b) when a is less than b:

$$R_4 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

R_4	1	2	3	4
1		×	×	×
2			×	×
3				×
4				

Conclusion: $(1, 2) \in R_4$ but $(2, 1) \notin R_4$; so R_4 is **not symmetric**.

Example 5: $R_5 = \{(a, b) \mid ab < 10\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_5 : A \to A = \{(a, b) \mid ab < 10\}$. Determine whether R_5 is symmetric.

Solution

List all (a, b) with $a \times b < 10$:

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,1), (4,2)\}.$$

R_5	1	2	3	4
1	×	×	×	×
2	×	×	×	×
3	×	×	×	
4	×	×		

Conclusion: Since $ab < 10$ implies $ba < 10$, whenever $(a, b) \in R_5$, $(b, a) \in R_5$ as well. Hence R_5 is **symmetric**.

Antisymmetric Relation

Definition

A relation R on a set A is called **antisymmetric** if and only if

$$(a, b) \in R \text{ and } (b, a) \in R \implies a = b.$$

- Both (a, b) and (b, a) can be in R **only when** $a = b$.
- In the table of the relation, if an off-diagonal entry (a, b) is marked, its mirror position (b, a) **must be empty**.

Example

If $A = \{1, 2, 3, 4\}$, the relation

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (3, 4)\}$$

is antisymmetric, because no distinct elements $a \neq b$ satisfy both (a, b) and (b, a) in R.

Example 1: $R_1 = \{(a, b) \mid a = b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_1 : A \to A = \{(a, b) \mid a = b\}$. Determine whether R_1 is antisymmetric.

Solution

All pairs have $a = b$:

$$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

R_1	1	2	3	4
1	$\times$			
2		$\times$		
3			$\times$	
4				$\times$

Conclusion: There are no pairs with $a \neq b$, so the antisymmetry condition holds trivially. Therefore R_1 is **antisymmetric**.

Example 2: $R_2 = \{(a, b) \mid a \neq b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_2 : A \to A = \{(a, b) \mid a \neq b\}$. Determine whether R_2 is antisymmetric.

Solution

All ordered pairs where a and b differ:

$$R_2 = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}.$$

R_2	1	2	3	4
1		×	×	×
2	×		×	×
3	×	×		×
4	×	×	×	

Conclusion: For example, both $(1, 2)$ and $(2, 1)$ exist. Since $1 \neq 2$, this violates antisymmetry. Hence R_2 is **not antisymmetric**.

Example 3: $R_3 = \{(a, b) \mid a \mid b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_3 : A \to A = \{(a, b) \mid a \text{ divides } b\}$. Determine whether R_3 is antisymmetric.

Solution

We include (a, b) when a divides b:

$$R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

R_3	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Conclusion: No pair of distinct a, b satisfies both (a, b) and (b, a). Therefore R_3 is **antisymmetric**.

Example 4: $R_4 = \{(a, b) \mid a < b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_4 : A \to A = \{(a, b) \mid a < b\}$. Determine whether R_4 is antisymmetric.

Solution

We include (a, b) when a is less than b:

$$R_4 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

R_4	1	2	3	4
1		×	×	×
2			×	×
3				×
4				

Conclusion: If $(a, b) \in R_4$, then $(b, a) \notin R_4$ because $b < a$ is false. Hence R_4 is **antisymmetric**.

Example 5: $R_5 = \{(a, b) \mid ab < 10\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_5 : A \to A = \{(a, b) \mid ab < 10\}$. Determine whether R_5 is antisymmetric.

Solution

List all (a, b) with $a \times b < 10$:

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,1), (4,2)\}.$$

R_5	1	2	3	4
1	×	×	×	×
2	×	×	×	×
3	×	×	×	
4	×	×		

Conclusion: Both $(1, 2)$ and $(2, 1)$ appear for distinct elements. Thus R_5 **is not antisymmetric**.

Transitive Relation

Definition

A relation R on a set A is called **transitive** if and only if

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R \quad \text{for all } a, b, c \in A.$$

- If a is related to b, and b is related to c, then a must also be related to c.
- In the table, whenever a path exists from $a \to b \to c$, the direct connection $a \to c$ must also exist.

Example

If $A = \{1, 2, 3, 4\}$, the relation

$$R = \{(1, 2), (2, 3), (1, 3)\}$$

is transitive because $(1, 2)$ and $(2, 3)$ imply $(1, 3)$, which is already in R.

Example 1: $R_1 = \{(a, b) \mid a = b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_1 : A \to A = \{(a, b) \mid a = b\}$. Determine whether R_1 is transitive.

Solution

$$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

R_1	1	2	3	4
1	$\times$			
2		$\times$		
3			$\times$	
4				$\times$

Conclusion: Whenever (a, b) and (b, c) occur, $a = b = c$, so (a, c) also occurs. Thus R_1 is **transitive**.

Example 2: $R_2 = \{(a, b) \mid a \neq b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_2 : A \to A = \{(a, b) \mid a \neq b\}$. Determine whether R_2 is transitive.

Solution

All ordered pairs where a and b differ:

$$R_2 = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}.$$

R_2	1	2	3	4
1		×	×	×
2	×		×	×
3	×	×		×
4	×	×	×	

Conclusion: Consider $(1, 2)$ and $(2, 1)$ — both are in R_2, but $(1, 1)$ is not (since $1 = 1$). This violates transitivity. Hence R_2 is **not transitive**.

Example 3: $R_3 = \{(a, b) \mid a \mid b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_3 : A \to A = \{(a, b) \mid a \text{ divides } b\}$. Determine whether R_3 is transitive.

Solution

We include (a, b) when a divides b:

$$R_3 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}.$$

R_3	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Conclusion: If $a \mid b$ and $b \mid c$, then $a \mid c$. This property always holds for divisibility. Therefore R_3 is **transitive**.

Example 4: $R_4 = \{(a, b) \mid a < b\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_4 : A \to A = \{(a, b) \mid a < b\}$. Determine whether R_4 is transitive.

Solution

We include (a, b) when a is less than b:

$$R_4 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

R_4	1	2	3	4
1		×	×	×
2			×	×
3				×
4				

Conclusion: If $a < b$ and $b < c$, then $a < c$ — the relation is always preserved. Hence R_4 is **transitive**.

Example 5: $R_5 = \{(a, b) \mid ab < 10\}$

Problem

Let $A = \{1, 2, 3, 4\}$ and define $R_5 : A \to A = \{(a, b) \mid ab < 10\}$. Determine whether R_5 is transitive.

Solution

List all (a, b) with $a \times b < 10$:

$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,1), (4,2)\}.$$

R_5	1	2	3	4
1	×	×	×	×
2	×	×	×	×
3	×	×	×	
4	×	×		

Conclusion: Take $(3, 2)$ and $(2, 4)$ — both satisfy $3 \times 2 < 10$ and $2 \times 4 < 10$, but $(3, 4)$ gives $3 \times 4 = 12 \not< 10$. Hence R_5 **is not transitive**.

Equivalence Relation

Definition

A relation R on a set A is called an **equivalence relation** if and only if it satisfies all three of the following properties:

(Reflexive) $(a, a) \in R$ for every $a \in A$.

(Symmetric) $(a, b) \in R \implies (b, a) \in R$.

(Transitive) $(a, b) \in R, (b, c) \in R \implies (a, c) \in R$.

- Every element is related to itself (reflexive).
- Relation works both ways (symmetric).
- "Chain" relationships remain consistent (transitive).

Intuition

Equivalence relations divide a set into distinct **equivalence classes** of elements that are "mutually related."

Equivalence Relation among R_1–R_5

Condition for Equivalence Relation

A relation R on a set A is an **equivalence relation** if and only if it is:

Reflexive, Symmetric, and Transitive.

Relation	Reflexive	Symmetric	Transitive	Equivalence?
$R_1 = \{(a,b) \mid a = b\}$	✓	✓	✓	**Yes**
$R_2 = \{(a,b) \mid a \neq b\}$	✗	✓	✗	No
$R_3 = \{(a,b) \mid a \mid b\}$	✓	✗	✓	No
$R_4 = \{(a,b) \mid a < b\}$	✗	✗	✓	No
$R_5 = \{(a,b) \mid ab < 10\}$	✗	✓	✗	No

Conclusion: Among all five relations, only R_1 $(a = b)$ is an **equivalence relation**.